Are Earnings Predictable?
Evidence From Equity Issues and Buyback Announcements[☆]

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Abstract

We find that market reactions to earnings announcements can be predictable. Four-factor abnormal returns to earnings announcements that follow buyback announcements are higher by 5.1% than similar returns to earnings announcements that follow equity issues over the (-1,+30) window; the difference is 2.2% when unadjusted returns are used. The magnitude is large and economically and statistically significant. The drift in these returns is unrelated and distinct from the post-earnings announcement drift. For example, we find positive drift for firms making buyback announcements even when they exhibit negative earnings surprises and find negative drift for firms issuing equity even when they show positive earnings surprises. Since the study looks at short periods around earnings announcements, it does not suffer from benchmarking errors that may influence long-horizon returns.

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1. Introduction

It is generally accepted that managers have more information about the firm than investors (Myers and Majluf, 1984; Miller and Rock, 1985; Healy and Palepu, 2001). Given this information asymmetry, managers can make informed decisions about corporate actions such as equity offerings, buybacks, dividends, insider trading, and mergers, in an effort to maximize value of the firm for current shareholders (Baker and Wurgler, 2002; Jenter, 2005; Dittmar and Thakor, 2007; Chan, Ikenberry, and Lee, 2007; DeAngelo, DeAngelo, and Stulz, 2010; Dong, Loncarski, Horst, and Veld, 2012; Dittmar and Field, 2015; Tarsalewska, 2018). Market participants evaluate these corporate actions to update their information about the firm and expected future firm performance.

In this paper, we examine whether information contained in corporate actions is fully incorporated into prices such that the market reaction to subsequent earnings announcements is unrelated to those corporate actions. In particular, we examine stock returns around earnings announcements that immediately follow these corporate actions noting that the next earnings announcement is probably a good indicator of the efficacy with which market participants reflect information disseminated through corporate actions.

This is in contrast to the vast literature in finance that has provided evidence of market inefficiency by documenting abnormal security returns over long observation periods (several months and years) following certain corporate events.[1] Instead of long-horizon periods, we adopt a short-horizon framework to abstract from potential benchmarking and other issues associated with long-horizon studies.[2] Our framework is designed to capture information revealed in the next corporate event, an earnings announcement. On average, earnings announcements should not generate any predictable market reaction provided information contained in prior corporate actions has been fully reflected in prices. However, if markets do not fully incorporate such information, they may react differentially and predictably based on the nature of the preceding event.

[1]These events include open market buybacks (Ikenberry, Lakonishok, and Vermaelen, 1995), seasoned equity offerings (Spiess and Affleck-Graves, 1995), initial public offerings (Ritter, 1991), dividend initiations and omissions (Michaely, Thaler, and Womack, 1995), stock splits (Ikenberry, Rankine, and Stice, 1996), mergers (Agrawal, Jaffe, and Mandelker, 1992), spinoffs (Cusatis, Miles, and Woolridge, 1993), tender offers (Lakonishok and Vermaelen, 1990), and short interest announcements (Boehmer, Huszar, and Jordan, 2010).

[2]Several recent studies show serious limitations of long-term return studies including the choice of benchmark models and matching algorithms. Kothari and Warner (1997, 2007) show that tests for multi-year abnormal performance around firm-specific events are severely misspecified and the lack of a perfect benchmark model of normal returns over horizons considered in long-term studies does not permit for reliable inferences.

We choose equity issues and buyback announcements as the corporate actions of interest for several reasons. First, managers can use their superior information to pick an appropriate time for issuing equity and announcing buybacks.[3] Second, the announcement of these corporate actions is voluntary and can be easily moved by a few weeks or months, unlike for mergers or asset sales where the announcement is a function of the status of negotiations between the contracting parties. Third, equity issuance and buybacks are two sides of the same coin (change in number of shares outstanding) so one can be considered a control group for the other. Finally, the frequency of equity offerings and buybacks, relative to other corporate actions, provides a large sample for analysis.

To test our hypothesis, we begin by examining the market reaction around earnings that follow buyback announcements and pricing of equity offerings. We find evidence of earnings predictability. First, we observe that there is incomplete adjustment to SEO pricings and buyback announcements that results in residual market reaction to earnings announcements. Second, we find that prices continue to drift after earnings announcements: upward for buybacks and downward for SEO pricings. More specifically, we find that the average raw return is +4.1% in a (-1,+30) window around earnings following buybacks closest to but 2 or more days prior to the earnings announcements compared with a raw return of 1.9% for SEO pricings, a difference of 2.2%. Similar results are observed for four-factor abnormal returns: a return of +1.7% around earnings following buyback announcements compared with -3.4% for SEO pricings over a (-1,+30) day window, a difference of 5.1%.[4] The return differentials are smaller for shorter windows, for example, a difference of 1.2% in raw returns and a difference of 2.7% in abnormal returns over the (+2,+15)

[3]For example, the studies of Taggart, Jr. (1977), Marsh (1982), Asquith and Mullins (1986), Korajczyk, Lucas, and McDonald (1991), Loughran, Ritter, and Rydqvist (1994), Jung, Kim, and Stulz (1996), Pagano, Panetta, and Zingales (1998), Hovakimian, Opler, and Titman (2001), and Eckbo, Masulis, and Norli (2007) show that the seasoned equity issues and initial public equity issues coincide with high valuations. Buybacks, on the other hand, coincide with low valuation, as shown in Ikenberry, Lakonishok, and Vermaelen (1995). Studies of long-run stock returns following corporate finance decisions show firms issue equity when the cost of equity is relatively low and repurchase equity when the cost is relatively high. See Stigler (1964), Ritter (1991), Loughran and Ritter (1995), Spiess and Affleck-Graves (1995), Brav and Gompers (1997), and Jegadeesh (2000) for more details. Studies of earning forecasts and realizations around equity issues show that firms tend to issue equity when investors are too enthusiastic about earnings prospects. Loughran and Ritter (1997), Rajan and Servaes (1997), Teoh, Welch, and Wong (1998a, 1998b), and Denis and Sarin (2001) are some sample studies in this category.

[4] The difference between raw and abnormal returns is unusually high for SEO issues. We reestimate abnormal returns using different platforms including Eventus, STATA, and SAS, and test a few observations with Excel, all of which confirm these results. We also test the abnormal returns using a three-factor model and a one-factor model. Using alternate periods for parameters estimation does not significantly alter the results. It seems that much of the four-factor adjustment arises from high values of systematic risk of firms issuing equity.

window. However, the returns remain economically and statistically significant.

The continuation of outperformance of buyback announcers and underperformance of equity issuers that we document here following earnings announcements is distinct and different from the well-documented post earnings announcement drift (PEAD) where stocks in the top and bottom deciles of returns continue to drift in the same direction. Unlike PEAD, the drift documented here does not depend on the initial market reaction to earnings announcement. Instead, we find a positive drift for earnings following buybacks and a negative drift for earnings following equity offerings *irrespective of the magnitude and sign* of first-day returns or earnings surprises. We examine this idea by forming deciles of earnings surprises and showing that abnormal returns around earnings announcements documented in this paper are independent of the magnitude and direction of earnings surprises. Also, in contrast to PEAD, regardless of whether earnings announcements contain positive, negative, or no surprises, we find that stock prices of firms with earnings preceded by buybacks tend to drift upward while stock prices of firms with earnings preceded by equity issuance tend to drift downward.

Among other explanations, we do not believe that the predictability of earnings reported here can be explained by confounding events including earnings management. The literature documents that managers have incentives to manipulate earnings prior to SEOs or buybacks which in turn affects returns around subsequent corporate events. For example, firms use accrual earnings management (Teoh, Welch, and Wong, 1998a,b; DuCharme, Malatesta, and Sefcik, 2004; Gong, Louis, and Sun, 2008; Cohen and Zarowin, 2010; Du and Shen, 2018), real earnings management (Kothari, Mizik, and Roychowdhury, 2016), and expectations management (Brockman, Khurana, and Martin, 2008) to influence equity market investors. We believe that the market reaction to SEO pricings and buyback announcements should have accounted for these systematic manipulations, leaving no surprises in earnings following corporate actions. Moreover, given the large samples used in this analysis, nonsystematic confounding events constitute white noise that should not affect the results. We also show that our results hold after 2002 in which the literature has found no evidence of earnings management for issuing and repurchasing firms (Fu and Haung, 2016).

We conduct several robustness tests and show that our main results are largely unaffected by alternate sample construction methodologies. In particular, we find that the results do not come from small stocks: the results are not materially affected when firms with market caps of less than $100 million are excluded. This is particularly important because long-term underperformance following equity issues is concentrated in the bottom tercile (Brav, 2000) or bottom quartile of firm size (Denis and Sarin, 2001). The results remain large and statistically significant over different subperiods.

We examine information asymmetry as an explanation for mispricing and return predictability following Korajczyk, Lucas, and McDonald (1991)'s finding that firms issue equity at times of low information asymmetry. We test this argument by using the average bid-ask spread, analysts' forecast error, and the standard deviation of analysts' forecasts as proxies for information asymmetry. Though we find that the average bid-ask spread is statistically significantly related to returns following SEO pricings and buyback announcements, we do not find broad support for information asymmetry as an explanation. Contrary to Korajczyk, Lucas, and McDonald (1991), we also find significant issuance of equity prior to earnings announcements when information asymmetry is likely to be elevated.[5]

Lie (2005), Gong, Louis, and Sun (2008), and Denis and Sarin (2001) examine abnormal returns over three days centered on earnings that occur a few quarters before and after buyback announcements and equity issues. However, they find evidence of earnings predictability only in a few quarters and not necessarily in the quarters immediately following corporate actions. The evidence presented in this paper is more compelling and systematic. The earlier papers also do not find a strong continuing drift that we document here. Separately and in addition, much of the prior literature has focused on long-term stock returns and operating performance of firms, which may be subject to benchmarking issues.[6] As assets are priced based on expected discounted cash flows, Cochrane (2011) notes that we will never be able to disentangle cash flow errors from changes

[5]In examining the nature of information conveyed by open-market buyback announcements, Bartov (1991) analyzes the surprise in actual earnings announcements relative to consensus forecasts but finds weak results. Instead, we focus on the market reaction to earnings announcements. We believe that the market's reaction to earnings announcement is a more appropriate tool than analyst forecasts to test our hypotheses because analyst forecasts have to be corrected for biases and those corrections themselves are likely to introduce new biases generating mixed results (Lim, 2001; Hong and Kubik, 2003).

[6]Several studies show that the long-term return methodologies are sensitive to the choice of benchmark. For example, Fama (1998) argues that most long-term return anomalies tend to disappear with reasonable changes in technique. Similarly, Kothari and Warner (1997), Barber and Lyon (1997), and Lyon, Barber, and Tsai (1999) show that using wrong benchmarks in measuring long-term abnormal returns would lead to erroneous inference on the significance of the event of interest.

in discount rates. In the case of issuing firms, Spiess and Affleck-Graves (1995), Loughran and Ritter (1995) and Brav, Geczy, and Gompers (2000) report underperformance of issuing firms over three-to-five years following the offering date. On the other hand, Eckbo, Masulis, and Norli (2000), Bessembinder and Zhang (2013), Li, Livdan, and Zhang (2009), and Bolton, Chen, and Wang (2013) show that these documented returns are due to incorrect firm matching, erroneous benchmarking, and changes in firm characteristics. Similarly, operating underperformance of issuing firms has been documented by Loughran and Ritter (1997) but disputed by Edelen, Ince, and Kadlec (2014). In the case of repurchasing firms, Ikenberry, Lakonishok, and Vermaelen (1995) and Peyer and Vermaelen (2009) show long-run excess returns whereas Fu and Haung (2016) cast doubt on the persistence of buyback anomalies. Lie (2005) documents that operating performance improves following buyback announcements but Grullon and Michaely (2004) find no evidence that repurchasing firms experience a considerable improvement in their operating performance relative to their peer firms.

Our study abstracts from the limitations of long-term studies and focuses instead on short windows around the earnings announcement immediately following buyback announcements and equity issues to provide a cleaner evaluation of mispricing and earnings predictability. The rest of this article is organized as follows: Section 2 describes the data, and summary statistics. Section 3 provides empirical results. Section 4 compares our results with the evidence related to post-earnings announcement drift, and Section 5 examines the robustness of main findings and implications. Finally, Section 6 explores the determinants of abnormal returns while Section 7 concludes.

2. Sample Selection and Summary Statistics

2.1. Data Sources

We obtain data on equity issuance and buybacks from Thomson Reuters SDC Platinum database, dates of earnings announcements from the Institutional Brokers Estimate System (IBES) database, and information on stock prices and returns from the Center for Research in Security Prices (CRSP) database. Compustat is used to calculate book value of equity for each firm. Firm-level data among different databases are matched using historical CUSIPs. We require firms included in the sample to have data on CRSP files at the time of announcement and have corresponding earnings announcement information on IBES.

Following Grullon and Michaely (2004), we include regulated firms (e.g. financial and utility firms) in the buyback and SEOs sample as they represent a significant fraction of the total sample. Also, following Fu and Haung (2016), we exclude buybacks by tender offers or through privately negotiated deals. In addition, to be consistent with Ikenberry, Lakonishok, and Vermaelen (1995), we analyze all open-market stock buyback announcements regardless of whether or not the programs were actually completed. These selection criteria generate a sample of 15,106 share buyback announcements between 1994 and 2015. Our sample of SEOs consists of new stock issues priced between 1970 and 2015, that are not IPOs. Following DeAngelo, DeAngelo, and Stulz (2010) and Fu (2010), we exclude SEOs made by closed-end funds, real estate investment trusts, and American Depository Receipts. This screening process generates a sample of 19,466 SEOs. Note that we use SEO pricing dates in our analyses which are different from SEO announcement dates.

2.2. Sample Statistics

The left hand side of Table A1 (in the Appendix) presents summary statistics for the sample of buyback announcements by year. The median market cap of a buyback firm is \$481 million, and firms announce a median buyback of 5.38% of shares outstanding.[7] The right hand side of Table A1 reports SEO pricings by year. The median percent of equity issues is 14.22% of shares outstanding.

Previous studies (Korajczyk, Lucas, and McDonald, 1991) argue that firms prefer to issue equity when the market is most informed about the quality of the firm. Hence, one should observe clustering of equity issues and buybacks after earnings releases. To visually examine this finding, Figure 1 plots the distribution of buybacks and SEOs around the closest quarterly earnings announcement.

Panel A of Figure 1 shows the frequency of stock buybacks and Panel B presents the frequency of equity pricings. These panels show that there is clustering of both SEO pricings and buyback announcements after earnings announcements. About 28% of buybacks and 32% of SEO pricings are announced during the first three weeks after earnings announcements date and this percentage decreases monotonically as we move further away from the earnings announcements date. However,

[7] All market capitalizations are in 2013 dollars, based on the All Urban Consumers (CPI-U) index from the U.S. Department of Labor, Bureau of Labor Statistics.

a considerable number of offerings take place before announcement of earnings. About 23% of stock buybacks and 17% of SEO pricings are made in the three weeks prior to earnings announcement.

Table A2 (in the Appendix) reports sample sizes used for different analyses in this paper. Panel A has the total number of buybacks and SEO pricings. Panel B reports the number of buybacks and SEO pricings that are closest to but prior to earnings announcements. Panel C shows the size of subsamples used for main analyses in the paper. These include all buybacks and SEO pricings that are closest to but 2 or more days prior to the earnings announcements. In constructing the observation windows, we maintain a minimum gap of 2 trading days between the start or end of each window and the announcement date of a corporate action so that there is no direct contamination of earnings by the corporate action. The results hold if we change the minimum gap to 5 days. Samples reported in Panels D and E are subsamples of Panel C in which small firms (market cap≤100 million) and financial and utility firms are excluded, respectively. Panel F is a subsample of Panel C obtained by imposing a minimum requirement of buyback or equity issuance (buybacks>5% and SEOs>10%). The minimum requirements are intended to exclude less important corporate actions. Finally, Panels G, H, I, and J show the size of subsamples for the subperiods.

3. Results

3.1. Returns Around Earnings Announcements

The main analysis in the paper relates to returns around earnings announcements that follow corporate actions to evaluate whether we can predict the nature of those earnings based on prior corporate actions. We acknowledge and recognize that the number of shares outstanding will decrease after buybacks that may cause the stock prices to increase. Similarly, the number of shares outstanding will increase after equity offerings that may cause stock prices to decrease. We expect the effect of such announcements to quickly get reflected in prices. In order not to contaminate announcements of subsequent earnings with the effect of buyback/equity offering announcements, we keep a gap of at least 2 business days between these corporate actions and earnings announcements.[8] As mentioned in the prior footnote, we also test our results with with wider gaps.

Results of our analysis are in Table 1 where the estimation of model parameters is based on the 251 to 31 day period prior to the event date. The left panel of Table 1 reports returns around earnings announcements for repurchasing firms that announce a buyback up to 2 days prior to earnings announcements. The repurchasing firms earn a positive, statistically significant abnormal return of 0.5% (t=4.602) from one day before announcement to one day after earnings announcement. The raw return over the same period is 0.80% (t=7.385). We also find a positive and statistically significant abnormal return of 1.1% (t=7.226) over a three-week period (+2,+15) after the announcement date and a return of 1.3% (t=6.068) over a six-week period (+2,+30). The abnormal returns increase to 1.7% (t=7.509) over the (-1,+30) window which includes announcement returns. The returns remain positive and statistically significant over other observation windows irrespective of whether they are reported as raw returns or four-factor abnormal returns.[9]

As for buybacks in the left panel, the middle panel reports abnormal returns around earnings announcements for issuing firms that priced SEOs up to 2 days before earnings announcements. The results show statistically negative returns around and after earnings announcements. For example, the four-factor abnormal return shows a loss of -1.6% (t=-13.383) over three weeks after the earnings announcement though the raw returns are positive (0.9%, see Footnote 4) and statistically significant.[10] The right panel shows the difference between returns around earnings announcements for repurchasing firms and issuing firms. The results show that the return differentials are statistically and economically significant over different observation windows. For instance, a portfolio that is long in buybacks and short in SEOs will produce raw returns of 1.2%, 1.7%, and 2.2% over (+2,+15), (+2,+30), and (-1,+30) windows, respectively. The portfolio will generate four-factor adjusted returns of 2.7%, 4.5%, and 5.1% over the same windows. Overall, the results show that

[8]We test the results with other windows. For example, we consider firms where the announcement of buybacks or SEO pricings occurs 6 to 15 days or 6 to 30 days before earnings and estimate the earnings announcement returns. The findings are consistent with the main results discussed in this section.

[9]In untabulated results, we also calculate the returns around the announcement of earnings for firms that announce a buyback program after earnings. We document significantly negative returns around the earnings announcement date. This finding implies that buybacks announced after an earnings announcement usually follow poor earnings.

[10]In untabulated results, we also calculate the returns around the announcement of earnings for firms that price an SEO after earnings. We document significantly positive returns around the earnings announcement date. This finding implies that SEOs priced after an earnings announcement usually follow good earnings.

earnings are predictable when they are preceded by buybacks or SEO pricings: positive market reaction to earnings preceded by buyback announcements and negative market reaction to earnings preceded by SEO pricings. The results reported in Table 1 are also shown in Figure 2.

4. Returns and Post Earnings Announcement Drift

In this section, we provide evidence that the continuation of returns around earnings following a buyback announcement or an equity offering that we document in this paper are distinct and are not a manifestation of the well-documented post earnings announcement drift (PEAD); the tendency for a stock to drift in the direction of an earnings surprise for several weeks following the earnings announcement. Based on the PEAD anomaly, estimated cumulative abnormal returns continue to drift up for good news firms and down for bad news firms even after earnings are announced. In contrast, we show that in our framework, stocks continue to drift up for earnings following buybacks regardless of whether earnings contain good news or bad news and continue to drift down for earnings following equity offerings regardless of the nature of the news contained in earnings. We begin our analysis by classifying stocks into deciles based on the observed raw returns around earnings over the (-1,+1) window and calculate the cross-sectional mean of raw and four-factor abnormal returns over (+2,+15) and (+2,+30) windows for each decile.

The results for the buybacks sample are in Table 2. The first row contains the decile of stocks with the lowest returns in the (-1,+1) window, with higher returns in subsequent rows. Looking at the (+2,+15) window after the earnings announcement, we note that raw returns and four-factor returns are similar across different deciles (from one decile to the next) irrespective of initial returns, and the difference between returns in the highest and lowest deciles is statistically insignificant as reported in the last row of Panel A. The results for the longer window, (+2,+30), are similar and the difference between returns in the highest and lowest deciles is statistically insignificant. The last column shows that firms in different deciles do not differ significantly in size, though the market-to-book ratio is much higher for the highest return decile.

As an additional test, in Panel B of Table 2, we classify stocks into two categories based on whether earnings following buybacks contain positive or negative surprises and calculate cross-sectional mean of raw and four-factor abnormal returns over (+2,+15) and (+2,+30) windows for each group. Earnings surprises are defined as the difference between the actual quarterly earnings per share and the median of all the latest outstanding forecasts (Bartov, Givoly, and Hayn, 2002; Akbas, 2016). We also report the cross-sectional mean of market-to-book and size for each group. Similar to the results in Panel A, the returns after earnings, calculated over (+2,+15) and (+2,+30) windows, are not statistically different across positive and negative earnings surprise portfolios.

Table 3 reports the results for SEOs sample. Similar to the results of previous analysis, we do not observe significant differences between returns after earnings across return decile portfolios or earnings surprise portfolios. Overall, the results in Tables 2 and 3 confirm that the drift in our sample is unrelated to the initial market reaction, and different from the nature of post-earnings announcement drift.

5. Robustness Tests

We conduct multiple robustness tests by altering the sample selection criteria. These tests support the results reported above. We begin by dropping the small size firms from our sample.

5.1. Excluding Small Size Firms

Prior empirical evidence of long-term abnormal performance has found that underperformance of issuing firms is concentrated among the smallest firms (Fama, 1998; Denis and Sarin, 2001). In addition, mean returns may be driven by small firms because those firms may experience large negative or positive returns in response to a corporate action. Such results are not meaningful because small firms form a small fraction of the overall size of the market and may also be illiquid. Therefore, as the first robustness check, we recalculate the returns around earnings announcements excluding firms with a market cap below $100 million. The results are presented in Panel A of Table 4. Similar to the main results presented in Table 1, we observe a positive market reaction to earnings announcements for firms that announce buybacks up to 2 days before earnings and a negative market reaction to earnings announcements for firms that price SEOs up to 2 days prior to earnings. These findings show that our results are not confined only to small firms which tend to show greater uncertainty regarding their value and to have higher trading costs. The return differentials reported in the last two columns of Panel A remain large and statistically significant after excluding small, high trading cost firms.

5.2. *Excluding Regulated Firms*

Regulated firms account for a significant fraction of our sample and were not excluded from the sample used to estimate the main results. Financial firms (SIC codes 6000–6999) may issue equity or repurchase their own stock to meet capital requirements and regulations. Similarly, utility firms' capital structure (SIC code 4900–4999) can be subject to regulatory supervision, which raises the possibility that observed SEOs and buybacks are a product of regulations. Therefore, we rerun our main analysis excluding firms in the regulated industries. The results are reported in Panel B of Table 4.

The sign and statistical significance of the results are unchanged. As earlier, the returns are positive and significant for firms that announce buybacks up to 2 days before earnings but are negative and significant for firms that price SEOs up to 2 days before announcing earnings. For example, over a three-week period (+2,+15), the mean abnormal return is significantly positive (1.4%, t=6.137) if the buyback announcement is made up to 2 days before announcement of earnings. For issuing firms, however, the abnormal return is significantly negative (-2.2%, t=-12.839) if the issue is priced up to 2 days before announcement of earnings. Compared to the results presented in Table 1, the return differentials are stronger when regulated firms are dropped from the sample. For example, a long-short portfolio will generate a raw return of 2.1% (t=5.179) and a four-factor adjusted return of 6.2% (t=15.015) over a (+2,+30) day window.

5.3. *Excluding Small Corporate Actions*

As another robustness test, we drop firms that announce a buyback of less than 5% or price an equity issuance of less than 10% of their shares outstanding. This test ensures that our results are not driven by less important corporate actions. The results are in Panel C of Table 4. We find the same pattern as we found for the full sample: positive and statistically significant market reaction to earnings following buybacks and negative and statistically significant market reaction to earnings following SEO pricings causing the predictability in earnings. The return differentials remain statistically and economically significant over different observation windows.

5.4. *Subperiod Analysis*

For the next robustness test, we split the sample into four subperiods for buyback and SEO samples to examine whether a particular period is responsible for the results. The subperiods are: 1994 to 2000, 2001 to 2005, 2006 to 2010, and 2011 to 2015. The results in Table 5 show that these periods exhibit similar earnings predictability but the results for the later periods are somewhat weaker though they continue to be statistically and economically significant. The weaker results over the recent periods are in line with the findings of Fu and Haung (2016) who document insignificant market reaction to announcements of buybacks and SEOs in the recent years. In contrast to their study, however, we study earnings announcements following buybacks and SEOs that allow us to capture market reaction over a subsequent informational event. Our findings show that earnings following corporate actions in recent years still surprise the market. The statistically and economically significant return differentials are present in both time periods though they are smaller in the second period.

There is a large literature on earnings management around stock buybacks (e.g., Gong, Louis, and Sun, 2008), and SEOs (e.g., Teoh, Welch, and Wong, 1998b; Rangan, 1998; Jo and Kim, 2007). These studies find a negative relation between earnings management and post-event stock return performance. For example, Gong, Louis, and Sun (2008) find that both post-buyback abnormal returns and reported improvement in operating performance are driven, at least in part, by pre-buyback downward earnings management rather than growth in profitability. It is thus likely that the abnormal performance following the post-buyback/issuance earnings announcements, documented in this paper, is due to earnings management. Fu and Haung (2016), however, find no evidence of earnings management after 2002 for repurchasing and issuing firms and relate this finding to the changing market environment including more efficient markets, and more transparent and credible information disclosure. Since our subperiod analysis shows that our results hold over periods with no or little earnings management, we believe our results are not due to earnings management prior to buybacks or equity issues.

5.5. *Returns Around Announcement of Corporate Actions (Buybacks and SEOs)*

Finally, we provide evidence that our sample is consistent with prior work by estimating returns around announcement of buybacks and SEOs and comparing them with the results documented in the literature. If a buyback program is announced, in line with prior work (e.g., Peyer and Vermaelen, 2009), we expect the stock price to move to a higher level and if an equity offering is priced, we expect the stock price to drop to a lower level (e.g., Jegadeesh, 2000; Denis and Sarin, 2001). To verify these prior findings, we create symmetric samples around the earnings announcement date in order to compare returns among samples with the same observation windows: the samples consist of firms that announce a buyback or SEO up to 2 days before the earnings announcement date. The results for both buybacks and SEOs are reported in Table 6.

The first two columns show the raw returns and four-factor abnormal returns around announcements of buybacks for various time intervals. Consistent with previous studies (Vermaelen, 1981; Chan, Ikenberry, and Lee, 2004; Peyer and Vermaelen, 2009), we find a positive market reaction to these announcements. For example, we find positive and statistically significant abnormal returns in the -1 to +1 window (2.0%, t=15.591) for firms that announced their buybacks up to 2 days before earnings announcements. The abnormal return increases to 2.6% (t=16.888) when the window is expanded to a (-1,+5) day period. The raw returns exhibit similar results confirming the findings of previous studies.

The right panel of Table 6 contains results for the issuing firms. Consistent with previous studies (Spiess and Affleck-Graves, 1995; Loughran and Ritter, 1995; Denis and Sarin, 2001), we find a negative market reaction to these announcements: the abnormal return in the -1 to +1 day period is a statistically significant -2.0% (t=-21.154) when equity issues are priced up to 2 days prior to earnings. The abnormal return continues to be negative: -2.0% (t=-17.265) in the (-1,+5) window. The raw returns show similar results. Overall, our sample is consistent with those used in previous studies demonstrating positive market reaction to buyback announcements and negative market reaction to equity offerings.

6. Regression Analysis

In this section, we examine firm characteristics that may explain the returns around earnings announcements documented above. Because information asymmetry may cause mispricing (Korajczyk, Lucas, and McDonald, 1991), the primary focus of this inquiry is on variables that measure information asymmetry.

6.1. Measures of Information Asymmetry

We use several factors that may contribute to a delay in market reaction to announcement of buybacks and equity issues. In particular, we consider the level of information asymmetry between managers and outside investors and construct three proxies to capture information asymmetry. The proxies include the bid-ask spread, analysts' forecast error, and the standard deviation of analysts' forecasts. The bid-ask spread is obtained from CRSP and the other proxies are obtained from IBES.

We select these proxies for information asymmetry based on prior research. Venkatesh and Chiang (1986) show that dealers increase bid-ask spreads with an increased degree of information asymmetry. They also find that the spreads significantly wider prior to earnings announcements. Given this finding, we calculate the bid-ask spread over 5 to 30 day before earnings announcement dates and use the average of the values as one of our measures of information asymmetry ($BidAskSpread$). Following Corwin and Schultz (2012), we estimate bid-ask spreads from daily high and low prices. They show that this measure outperforms the covariance spread estimator of Roll (1984) and the estimator of Lesmond, Ogden, and Trzcinka (1999). Elton, Gruber, and Gultekin (1984) provide evidence on the use of errors in analysts' forecasts of earnings and the standard deviation of analysts' forecasts as measures of information asymmetry, that is, firms with higher levels of information asymmetry between managers and outside investors are expected to have higher forecast errors and higher standard deviation of the forecasts submitted by analysts.

For the accuracy of analysts' forecasts of earnings per share ($ForecastError$) and the dispersion among analysts' forecasts ($StdForecast$) as proxies for information asymmetry, we follow

Krishnaswami and Subramaniam (1999) and define them as:

$$ForecastError = \left| \frac{E_{i,q} - \overline{E}_{i,q}}{P_{i,q}} \right|, \text{ and} \tag{1}$$

$$StdForecast = \left| \frac{Std_{i,q}}{P_{i,q}} \right|, \tag{2}$$

where $E_{i,q}$ is the actual quarterly earnings per share for firm i in quarter q, $\overline{E}_{i,q}$ is the average of all the latest outstanding forecasts (among those less than 90 days old), $Std_{i,q}$ is the standard deviation of all the latest outstanding earnings forecasts and $P_{i,q}$ is the share price at the beginning of the quarter.

6.2. Control Variables

Besides firm size, market-to-book ratio, prior month firm return, and prior month market return, we control for the frequency of buybacks and equity issues. Jagannathan and Stephens (2003) define an "infrequent" buyback or SEO as the first announcement in a five-year period, a moderate buyback or SEO as the second announcement in a five-year period, and a "frequent" buyback or SEO as the third or higher announcement in a five-year period. We include this proxy in our specification as a dummy variable which takes a value of 1 for moderate and frequent announcements. We control for the frequency of buybacks and equity issues because prior studies (e.g., Dittmar and Field, 2015) show that frequent and infrequent repurchasers (issuers) may experience different market reactions following the announcement of corporate actions. Table 7 reports descriptive statistics of the regressors.

An examination of Panel A reveals that forecast error for the repurchasing firms, on average, is 0.57% of their stock price and the standard deviation of the forecasts is 0.21% of their stock price. Panel A also shows that 32% of the repurchasing firms are moderate or frequent repurchasers and they are, on average, large firms (average size of $6.844 billion). Panel B, on the other hand, shows that forecast error for the issuing firms, on average, is 5.48% of their stock price and the standard deviation of the forecasts is 1.60% of their stock price. Panel B also shows that 42% of the issuing firms are moderate or frequent issuers and they are smaller than the repurchasing firms (average size of $2.014 billion). The difference in firm size is not unexpected because larger firms are usually mature and may generate excess cash for distribution whereas smaller firms need equity to grow. Repurchasing firms have a mean return of -4.24% in the month prior to the buyback announcement pointing to a possible undervaluation at the time of the announcement. Similarly, equity issues are priced after the stock price has gone up by 1.58% in the previous month pointing to a possible overvaluation at the time that the equity issue is priced.

6.3. Regression Models

To analyze the relation between returns around earnings and firm characteristics, we estimate the following cross-sectional regression for repurchasing and issuing firms separately:

$$
\begin{aligned}
R_i(\tau_1, \tau_2) = \beta_0 &+ \beta_1 AsymProxy_i + \beta_2 Log(Size)_i + \beta_3 FirmRet_i \\
&+ \beta_4 MktRet + \beta_5 (M/B)_i + \beta_6 FrequentDum_i + \varepsilon_i,
\end{aligned}
\tag{3}
$$

where $R_i(\tau_1, \tau_2)$ is the return for firm i from time τ_1 to τ_2 around earnings announcements and $AsymProxy_i$ represents our proxy for information asymmetry between firm i's managers and investors. The proxies include $BidAskSpread_i$, $Log(1 + ForecastError)_i$, and $Log(1 + StdForecast)_i$ which we include both individually and collectively in our baseline regression. $Log(Size)_i$ is the logarithm of market capitalization for firm i, $FirmRet_i$ is the cumulative return over a 1-month period before the buyback announcement or SEO pricing for firm i, $MktRet$ is the cumulative market return over the same period, M/B_i is the ratio of the market value to the book value of equity for firm i, and $FrequentDum_i$ is a dummy variable which takes a value of 1 for moderate and frequent buybacks or SEOs. To mitigate the effect of outliers, $ForecastError$, $StdForecast$ and M/B are winsorized at the 1% and 99% of their empirical distribution.

We choose four-factor abnormal returns R_i(+2,+15) and R_i(+2,+30) as dependent variables in this setup and run the regression for firms that make their buyback announcements or SEO pricings up to 2 days before their closest quarterly earnings. We add size to control for any differences in market reaction to earnings surprises of large firms versus earnings surprises of small firms. We add firm and market returns to the model to control for prior price movements in the market and the firm. In addition, we control for market-to-book ratio to detect differences in market reaction to the earnings of growth firms versus value firms. The estimates for repurchasing firms are in Table 8 and for issuing firms are in Table 9.

Models 1 to 4 in Table 8 contain the regression estimates for Equation (3) if a buyback announcement occurs up to 2 days before the earnings announcement and four-factor abnormal return $R_i(+2,+15)$ is used as the dependent variable. In the first specification, we use the bid-ask spread as a proxy for information asymmetry while for the second and third specifications we use analysts' forecast error and the standard deviation of analysts' forecasts as the proxies of information asymmetry. In the fourth specification, we include all the information asymmetry proxies in our baseline model. The coefficients on the bid-ask spread in Models 1 and 4 are significantly and positively related to earnings abnormal returns. Assuming that dealers widen the bid-ask spread due to greater information asymmetry prior to earnings announcements that usually convey new information (Venkatesh and Chiang, 1986), the positive coefficient suggests that the abnormal returns are higher when information asymmetry is higher.

The other proxies for information asymmetry are statistically insignificant. Average past firm returns ($FirmRet$) are negatively related to returns after earnings announcements implying that market is less surprised by earnings of the repurchasing firms that have been enjoying positive price movements before earnings announcements. Over this window, past market return, market-to-book ratio, and being a moderate or frequent repurchaser are statistically insignificant. In Models 5 to 8, we analyze the returns over a six-week (+2,+30) window. The coefficients on the bid-ask spread in Model 5 and 8 and the coefficient on the standard deviation of analysts' forecasts in Model 7 are positive and statistically significant at 1% and 5%, respectively. The remaining control variables excluding market-to-book ratio show similar patterns as in previous models. In summary, the findings of Table 8 provide evidence that earnings abnormal returns preceded by buyback announcements are partially caused by the level of information asymmetry when proxied by the average bid-ask spread. This suggests that market reaction to the announcement of buybacks is incomplete and the following earnings announcement continues to surprise the market.

For the SEOs sample, we implement similar models as above to investigate the relation between abnormal returns and firm characteristics. Table 9 contains the regression results for Equation (3) when the SEO is priced up to 2 days before earnings. Models 1 to 4 in Table 9 include the regression estimates using four-factor abnormal return $R_i(+2,+15)$ as the dependent variable. In the first three models in which the information asymmetry proxies enter the model individually, the average bid-ask spread and analysts' forecast error are negatively and significantly associated with earnings abnormal returns. In Model 4 in which all the information asymmetry proxies are included, the average bid-ask spread continues to be statistically significant with the expected sign. Recall that abnormal returns around earnings following equity pricings are negative. Hence, the negative estimate on the information asymmetry proxies suggests that abnormal returns around earnings are more negative when the level of information asymmetry is higher. Over the (+2,+15) window, past market performance and market-to-book ratio are statistically insignificant and have no sizable impact on abnormal returns. The results also show that market is less surprised by earnings of moderately or frequently issuing firms. Analyzing the returns over a six-week (+2,+30) window also leads to the same conclusions: higher information asymmetry between managers and outside investors leads to higher abnormal returns around and after earnings announcements. In summary, the findings of Table 9 provide evidence that market reaction to SEO pricings is incomplete which in turn causes predictability in earnings that immediately follow these SEOs.

7. Earnings Predictability and Conclusions

7.1. Earnings Predictability

Assuming no predictability, the average market reaction to a large sample of earnings announcements should not be significantly different from zero. However, as reported above, the market reaction to earnings following buybacks is positive and statistically significant: the (-1,+1) day raw return is 0.8%, and the abnormal return is 0.5%. If the window is expanded to a longer period, the returns become larger: raw returns of 4.1% and 3.3% and abnormal returns of 1.7% and 1.3% over (-1,+30) and (+2,+30) windows, respectively, are observed. On the other hand, the market reaction to earnings after equity pricings is slightly weaker. The raw return over a (-1,+1) window is 0.4% and the abnormal return is -0.2% and statistically significant. As we expand the window, the returns become larger: raw returns of 1.9% and 1.6% and abnormal returns of -3.4% and -3.2% over (-1,+30) and (+2,+30) windows, respectively, with all returns being statistically significant. If we consider the longest observation window reported, (-1,+30), the market reaction to earnings announcements following buyback announcements is 2.2% greater than the reaction to earnings announcements following equity pricings when raw returns are considered, and 5.1% when abnormal returns are considered. Both returns are statistically and economically significant. From the difference in market reactions, we can conclude that market adjustment to corporate actions is incomplete, and can result in predictability of earnings announcements.

7.2. Conclusion

Consistent with prior work, we find that stock issues are accompanied by a negative market reaction while buyback announcements are accompanied by a positive market reaction. While these corporate actions reduce information asymmetry, we find that the market reaction to these signals is incomplete which leads to predictable returns around earnings announcements such that stock returns around the next earnings announcement are systematically greater following buyback announcements than following equity pricings. As mentioned above, these returns are statistically and economically significant. We show that the excess returns are not related to post-earnings announcement drift as that drift is conditioned on an extreme reaction to an earnings announcement, whereas the results in this paper point to predictability of returns for buyback announcements and equity issues irrespective of the magnitude and direction of that reaction prior to announcement of earnings.

Given the high frequency of buybacks and equity issues, there should not be any residual predictability. We examine reasons for incomplete market reaction to corporate actions and find that the results are not driven by small, illiquid firms as the results are not materially affected when firms with a market cap below $100 million are excluded nor are they driven by any other known subset of stocks that may cause such predictability. Though earnings predictability documented in this study may be partially related to information asymmetry, a significant portion of predictability can be attributed to the inability of market participants to fully incorporate information contained in prior corporate actions.

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Table 1: Returns around earnings announcement dates

This table reports returns around earnings announcement dates for both repurchasing and issuing firms. The analysis is conducted for firms that announce buybacks or price an equity offering up to 2 days prior to the next earnings announcement. In addition to raw returns, value-weighted four-factor (Fama-French-Carhart) abnormal returns are reported. The last two columns show the difference between returns around earnings announcement dates for repurchasing firms and issuing firms. The t statistics are reported in parentheses below each estimate. N shows the number of observations. The methodology used to compute the CARs for the four factor model is as follows. Assume an asset pricing model: $R_{it} = \alpha_i + \beta_{im}R_{mt} + \beta_{is}SMB_t + \beta_{ih}HML_t + \beta_{iu}UMD_t + \epsilon_{it}$. An estimation period from day 251 to day 31 before event date is used to calculate the parameters of the model provided there are at least 15 daily returns available for estimation. The CAR at time τ_2 relative to time τ_1 is then computed as:

$$CAR_{(\tau_1,\tau_2)} = \frac{1}{n}\sum_{\tau=\tau_1}^{\tau_2}\sum_{i=1}^{n}(R_{it} - \hat{\alpha}_i - \hat{\beta}_{im}R_{mt} - \hat{\beta}_{is}SMB_t - \hat{\beta}_{ih}HML_t - \hat{\beta}_{iu}UMD_t)$$

Day 0: Earnings announcement date						
	Buybacks (Up to 2 days before earnings)		SEOs (Up to 2 days before earnings)		Buybacks - SEOs (Difference)	
Window	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw
(-1,+1)	0.005	0.008	−0.002	0.004	0.007	0.004
	(4.602)	(7.385)	(−2.763)	(5.147)	(5.362)	(3.243)
(0,+1)	0.003	0.005	−0.002	0.001	0.006	0.004
	(3.351)	(5.246)	(−3.800)	(2.164)	(4.864)	(3.267)
(-1,+3)	0.006	0.011	−0.004	0.005	0.011	0.007
	(5.278)	(9.316)	(−5.351)	(5.268)	(7.393)	(4.551)
(-1,+5)	0.008	0.015	−0.007	0.005	0.015	0.009
	(6.157)	(11.177)	(−7.461)	(5.589)	(9.373)	(5.712)
(-1,+30)	0.017	0.041	−0.034	0.019	0.051	0.022
	(7.509)	(17.691)	(−16.940)	(10.355)	(16.767)	(7.232)
(+2,+15)	0.011	0.021	−0.016	0.009	0.027	0.012
	(7.226)	(14.240)	(−13.383)	(7.488)	(14.083)	(6.464)
(+2,+30)	0.013	0.033	−0.032	0.016	0.045	0.017
	(6.068)	(16.188)	(−17.323)	(9.112)	(16.061)	(6.497)
N	4,737	4,737	8,846	8,846		

Table 2: Return deciles: Buybacks sample

Return deciles based on earnings announcement returns are formed and reported in Panel A for different windows. The last two columns report the average of market-to-book ratio and the average size of firms in each portfolio. Panel B reports the same values as Panel A for portfolios formed based on whether earnings following buybacks contain positive or negative surprises. Earnings surprises are defined in Section 4. The analysis is done for the firms that announce buybacks up to 2 days before their earnings release date. The *t* statistics are reported in parentheses below each estimate. N shows the number of observations. See Table 1 for the methodology used for calculating abnormal returns.

	Window: (-1,+1)		Window: (+2,+15)		Window: (+2,+30)			
	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	M/B	Size
Panel A: Return deciles								
Decile 1	−0.121	−0.108	0.031	0.020	0.047	0.029	2.432	5916.485
	(−31.515)	(−24.733)	(4.433)	(3.165)	(5.311)	(3.188)	(15.979)	(4.023)
Decile 2	−0.044	−0.040	0.007	0.002	0.025	0.009	2.323	11727.717
	(−91.254)	(−24.717)	(1.238)	(0.347)	(3.287)	(1.273)	(16.646)	(4.766)
Decile 3	−0.023	−0.022	0.025	0.018	0.032	0.020	2.275	10173.356
	(−91.444)	(−17.960)	(5.096)	(4.116)	(4.678)	(3.095)	(15.728)	(4.297)
Decile 4	−0.009	−0.010	0.008	0.005	0.012	0.003	2.507	14483.143
	(−45.238)	(−7.705)	(1.933)	(1.220)	(1.787)	(0.465)	(13.096)	(4.205)
Decile 5	0.000	0.000	0.019	0.013	0.030	0.016	2.257	7968.353
	(−0.366)	(−0.016)	(3.354)	(2.194)	(3.804)	(1.967)	(12.773)	(2.825)
Decile 6	0.009	0.005	0.021	0.011	0.029	0.008	2.345	8653.240
	(44.294)	(4.834)	(4.605)	(2.530)	(4.473)	(1.170)	(15.136)	(4.610)
Decile 7	0.021	0.016	0.022	0.012	0.030	0.012	2.451	9204.068
	(97.171)	(15.034)	(5.722)	(3.149)	(4.776)	(1.864)	(17.166)	(4.875)
Decile 8	0.035	0.029	0.022	0.014	0.044	0.021	2.438	11627.586
	(122.431)	(20.110)	(4.569)	(3.031)	(5.954)	(2.823)	(16.374)	(4.571)
Decile 9	0.061	0.051	0.028	0.017	0.046	0.024	2.928	12678.662
	(101.334)	(34.135)	(5.448)	(3.356)	(5.828)	(2.905)	(15.788)	(3.955)
Decile 10	0.147	0.127	0.042	0.019	0.058	0.024	3.316	3395.852
	(35.621)	(28.363)	(5.754)	(2.915)	(5.446)	(2.406)	(14.968)	(4.323)
10-1	0.268	0.235	0.011	−0.001	0.011	−0.005	0.884	−2520.632
	(47.547)	(37.567)	(1.132)	(−0.106)	(0.760)	(−0.350)	(3.289)	(−1.512)
Panel B: Earnings surprise portfolios								
Negative (33%)	−0.009	−0.009	0.023	0.016	0.033	0.024	2.202	5340.703
	(−3.637)	(−3.888)	(7.728)	(5.810)	(8.301)	(5.606)	(26.920)	(6.474)
Positive (67%)	0.015	0.011	0.023	0.012	0.037	0.014	2.706	11571.203
	(8.814)	(6.665)	(10.545)	(5.909)	(11.739)	(4.544)	(39.508)	(11.202)
Pos-Neg	0.024	0.020	0.000	−0.004	0.004	−0.010	0.504	6230.499
	(8.136)	(7.052)	(0.117)	(−1.276)	(0.757)	(−1.854)	(4.720)	(4.713)
N	4,737	4,737						

Table 3: Return deciles: SEOs sample

Return deciles based on earnings announcement returns are formed and reported in Panel A for different windows. The last two columns report the average of market-to-book ratio and the average size of firms in each portfolio. Panel B reports the same values as Panel A for portfolios formed based on whether earnings following buybacks contain positive or negative surprises. Earnings surprises are defined in Section 4. The analysis is done for the firms that price an equity offering up to 2 days before their earnings release date. The *t* statistics are reported in parentheses below each estimate. N shows the number of observations. See Table 1 for the methodology used for calculating abnormal returns.

	Window: (-1,+1)		Window: (+2,+15)		Window: (+2,+30)			
	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	M/B	Size
Panel A: Return deciles								
Decile 1	−0.121	−0.107	0.004	−0.033	0.006	−0.071	4.237	1683.804
	(−36.552)	(−30.120)	(0.626)	(−5.378)	(0.595)	(−6.774)	(16.350)	(4.273)
Decile 2	−0.045	−0.043	−0.003	−0.026	−0.007	−0.056	3.615	2632.007
	(−108.358)	(−32.417)	(−0.590)	(−5.224)	(−0.902)	(−6.986)	(16.393)	(4.630)
Decile 3	−0.024	−0.025	0.005	−0.021	0.007	−0.043	3.110	2072.810
	(−111.878)	(−20.540)	(0.822)	(−3.558)	(0.846)	(−4.782)	(16.580)	(8.945)
Decile 4	−0.011	−0.013	0.003	−0.018	0.009	−0.034	3.107	2438.642
	(−64.169)	(−10.698)	(0.658)	(−3.639)	(1.247)	(−4.632)	(14.803)	(6.377)
Decile 5	−0.002	−0.007	−0.001	−0.017	0.011	−0.031	2.601	2799.720
	(−13.714)	(−5.926)	(−0.100)	(−3.612)	(1.795)	(−4.350)	(16.538)	(4.523)
Decile 6	0.005	0.000	0.008	−0.014	0.017	−0.027	2.550	4352.678
	(35.572)	(0.067)	(2.050)	(−3.299)	(2.608)	(−3.739)	(19.804)	(3.290)
Decile 7	0.016	0.008	0.013	−0.009	0.021	−0.034	3.510	1936.442
	(92.356)	(7.419)	(2.613)	(−1.917)	(2.924)	(−4.517)	(16.091)	(7.808)
Decile 8	0.029	0.019	0.000	−0.027	0.010	−0.044	3.539	2473.909
	(119.727)	(18.034)	(−0.070)	(−5.512)	(1.448)	(−5.920)	(15.042)	(3.571)
Decile 9	0.054	0.041	0.007	−0.020	0.014	−0.050	3.837	2598.112
	(121.255)	(27.765)	(1.130)	(−3.457)	(1.578)	(−5.313)	(16.411)	(6.366)
Decile 10	0.135	0.108	0.010	−0.034	0.021	−0.066	4.870	1739.901
	(40.637)	(31.583)	(1.531)	(−4.984)	(2.141)	(−5.624)	(17.275)	(6.196)
10-1	0.256	0.215	0.006	−0.002	0.015	0.005	0.633	56.097
	(54.590)	(43.608)	(0.654)	(−0.185)	(1.078)	(0.309)	(1.654)	(0.116)
Panel B: Earnings surprise portfolios								
Negative (37%)	−0.009	−0.012	0.005	−0.017	0.009	−0.036	3.476	2237.175
	(−4.585)	(−7.161)	(1.588)	(−5.758)	(2.079)	(−7.464)	(26.457)	(7.275)
Positive (63%)	0.009	0.003	0.005	−0.026	0.011	−0.053	3.623	2512.647
	(6.159)	(1.965)	(2.142)	(−11.652)	(3.487)	(−14.996)	(41.873)	(11.857)
Pos-Neg	0.018	0.015	0.000	−0.008	0.002	−0.017	0.147	275.472
	(7.441)	(6.814)	(−0.059)	(−2.294)	(0.322)	(−2.875)	(0.932)	(0.738)
N	8,846	8,846						

Table 4: Returns around earnings announcement dates: Robustness tests

This table reports returns around earnings announcement dates for both repurchasing and issuing firms. The analysis is conducted for firms that announce buybacks or price an equity offering up to 2 days prior to the next earnings announcement. Panel A reports the returns around earnings announcements for both repurchasing and issuing firms excluding small firms with a market cap below $100 million. Panel B reports the returns around earnings announcements excluding financial and utility firms. Panel C reports the returns around earnings announcement dates excluding firms that announce a buyback of less than 5% or announce an equity issuance of less than 10% of their shares outstanding. The last two columns in each panel show the difference between returns around earnings announcement dates for repurchasing firms and issuing firms. In addition to raw returns, four-factor (Fama-French-Carhart) abnormal returns are reported in all panels. The *t* statistics are reported in parentheses below each estimate. N shows the number of observations. See Table 1 for the methodology used for calculating abnormal returns.

Day 0: Earnings announcement date						
	Buybacks (Up to 2 days before earnings)		SEOs (Up to 2 days before earnings)		Buybacks - SEOs (Difference)	
Window	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw
Panel A: Excluding small firms						
(-1,+1)	0.004	0.007	−0.001	0.005	0.005	0.002
	(3.235)	(5.410)	(−1.590)	(6.477)	(3.594)	(1.248)
(0,+1)	0.003	0.005	−0.001	0.002	0.004	0.002
	(2.537)	(3.968)	(−2.249)	(3.768)	(3.314)	(1.595)
(-1,+3)	0.006	0.011	−0.005	0.005	0.010	0.006
	(4.435)	(7.740)	(−5.335)	(5.619)	(6.618)	(3.378)
(-1,+5)	0.008	0.015	−0.007	0.007	0.015	0.008
	(5.728)	(9.628)	(−6.990)	(6.469)	(8.653)	(4.342)
(-1,+30)	0.017	0.039	−0.036	0.021	0.053	0.018
	(6.929)	(15.490)	(−17.311)	(10.919)	(16.456)	(5.560)
(+2,+15)	0.010	0.020	−0.017	0.010	0.027	0.010
	(6.977)	(12.821)	(−14.344)	(8.106)	(14.469)	(5.277)
(+2,+30)	0.013	0.032	−0.035	0.016	0.048	0.016
	(6.078)	(14.723)	(−18.340)	(9.236)	(16.665)	(5.638)
N	3,407	3,407	7,299	7,299		
Panel B: Excluding regulated firms						
(-1,+1)	0.007	0.011	−0.002	0.004	0.009	0.007
	(4.311)	(6.408)	(−2.265)	(4.018)	(4.861)	(3.340)
(0,+1)	0.004	0.007	−0.003	0.001	0.007	0.006
	(2.931)	(4.549)	(−3.054)	(1.638)	(4.066)	(3.113)
(-1,+3)	0.009	0.015	−0.005	0.005	0.014	0.010
	(4.993)	(7.723)	(−4.560)	(4.169)	(6.665)	(4.209)
(-1,+5)	0.010	0.018	−0.008	0.006	0.019	0.012
	(5.090)	(8.716)	(−6.396)	(4.531)	(7.789)	(4.718)
(-1,+30)	0.026	0.049	−0.046	0.021	0.072	0.027
	(7.261)	(13.751)	(−15.837)	(7.950)	(15.606)	(6.113)
(+2,+15)	0.014	0.024	−0.022	0.009	0.036	0.015
	(6.137)	(10.475)	(−12.839)	(5.527)	(12.700)	(5.108)
(+2,+30)	0.019	0.038	−0.043	0.017	0.062	0.021
	(5.929)	(12.196)	(−16.285)	(6.956)	(15.015)	(5.179)
N	2,760	2,760	5,793	5,793		

Continued on next page

Table 4: Continued from previous page

Panel C: Excluding buybacks<5% and SEOs<10%						
(-1,+1)	0.007	0.009	−0.002	0.004	0.009	0.005
	(4.839)	(6.441)	(−2.066)	(4.182)	(5.197)	(3.154)
(0,+1)	0.005	0.006	−0.002	0.001	0.007	0.005
	(3.602)	(4.804)	(−3.077)	(1.720)	(4.671)	(3.211)
(-1,+3)	0.009	0.013	−0.005	0.005	0.013	0.009
	(5.423)	(8.147)	(−4.394)	(4.125)	(6.947)	(4.464)
(-1,+5)	0.011	0.018	−0.008	0.005	0.019	0.013
	(6.465)	(10.145)	(−6.414)	(3.934)	(8.962)	(6.076)
(-1,+30)	0.024	0.046	−0.038	0.018	0.063	0.027
	(7.740)	(14.305)	(−14.918)	(7.619)	(15.437)	(6.860)
(+2,+15)	0.015	0.026	−0.019	0.007	0.034	0.018
	(7.467)	(12.232)	(−12.451)	(4.830)	(13.471)	(7.134)
(+2,+30)	0.017	0.036	−0.037	0.014	0.054	0.022
	(6.206)	(12.876)	(−15.306)	(6.518)	(14.642)	(6.131)
N	2,741	2,741	5,976	5,976		

Table 5: Returns around earnings announcement dates: Subperiod analysis

This table reports the returns around earnings announcements for both repurchasing and issuing firms for four subperiods: 1994 to 2000, 2001 to 2005, 2006 to 2010, and 2011 to 2015. The analysis is conducted for firms that announce buybacks or price SEOs up to 2 days prior to the next earnings announcement. In addition to raw returns, four-factor (Fama-French-Carhart) abnormal returns are reported. The last four columns show the difference between returns around earnings announcement dates for repurchasing firms and issuing firms. The *t* statistics are reported in parentheses below each estimate. N shows the number of observations. See Table 1 for the methodology used for calculating abnormal returns.

Day 0: Earnings announcement date												
	Buybacks (Up to 2 days before earnings)				SEOs (Up to 2 days before earnings)				Buybacks - SEOs (Difference)			
	1994 to 2000		2001 to 2005		1994 to 2000		2001 to 2005		1994 to 2000		2001 to 2005	
Window	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw
(-1,+1)	0.005	0.008	0.007	0.013	−0.001	0.008	−0.003	0.001	0.006	0.000	0.010	0.012
	(3.521)	(5.818)	(2.692)	(4.908)	(−0.393)	(4.067)	(−1.497)	(0.459)	(2.361)	(−0.112)	(3.011)	(3.412)
(0,+1)	0.003	0.005	0.005	0.009	−0.001	0.004	−0.005	−0.002	0.004	0.001	0.010	0.011
	(2.447)	(4.291)	(2.226)	(3.591)	(−0.852)	(2.465)	(−2.262)	(−0.884)	(2.104)	(0.444)	(3.160)	(3.331)
(-1,+3)	0.005	0.011	0.009	0.018	−0.008	0.008	−0.007	0.000	0.013	0.003	0.016	0.019
	(3.582)	(7.002)	(3.087)	(5.978)	(−3.408)	(3.333)	(−2.614)	(−0.147)	(4.829)	(1.136)	(4.043)	(4.554)
(-1,+5)	0.007	0.015	0.010	0.023	−0.014	0.007	−0.008	0.004	0.022	0.008	0.018	0.019
	(4.256)	(8.264)	(3.221)	(7.298)	(−5.676)	(2.720)	(−2.820)	(1.246)	(7.087)	(2.387)	(4.271)	(4.380)
(-1,+30)	0.019	0.040	0.017	0.069	−0.081	0.010	−0.034	0.024	0.100	0.030	0.051	0.046
	(5.959)	(12.427)	(3.011)	(12.518)	(−13.766)	(1.856)	(−5.400)	(4.023)	(14.940)	(4.864)	(6.043)	(5.629)
(+2,+15)	0.012	0.024	0.008	0.031	−0.042	0.001	−0.015	0.011	0.054	0.023	0.024	0.020
	(6.047)	(11.688)	(2.690)	(9.807)	(−12.574)	(0.419)	(−4.245)	(3.035)	(13.884)	(6.131)	(4.970)	(4.036)
(+2,+30)	0.014	0.032	0.010	0.056	−0.080	0.002	−0.031	0.023	0.095	0.030	0.040	0.034
	(4.846)	(11.103)	(2.060)	(11.917)	(−14.739)	(0.320)	(−5.439)	(4.199)	(15.267)	(5.391)	(5.482)	(4.674)
N	2,434	2,434	1,040	1,040	1,679	1,679	815	815				
	2006 to 2010		2011 to 2015		2006 to 2010		2011 to 2015		2006 to 2010		2011 to 2015	
Window	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw	Four-factor	Raw
(-1,+1)	0.005	0.004	0.002	0.004	−0.007	−0.002	−0.009	−0.006	0.012	0.006	0.011	0.010
	(1.539)	(1.249)	(0.588)	(1.343)	(−2.862)	(−0.708)	(−4.536)	(−2.977)	(3.014)	(1.411)	(3.150)	(2.842)
(0,+1)	0.003	0.003	0.001	0.002	−0.007	−0.003	−0.007	−0.005	0.010	0.006	0.007	0.006
	(1.140)	(0.867)	(0.279)	(0.587)	(−3.099)	(−1.469)	(−3.650)	(−2.539)	(2.804)	(1.595)	(2.344)	(1.937)
(-1,+3)	0.006	0.007	0.004	0.006	−0.010	−0.003	−0.010	−0.006	0.016	0.010	0.014	0.012
	(1.913)	(1.941)	(1.441)	(2.056)	(−3.179)	(−0.885)	(−4.265)	(−2.417)	(3.574)	(2.009)	(3.765)	(3.116)
(-1,+5)	0.008	0.009	0.006	0.009	−0.011	−0.004	−0.015	−0.010	0.020	0.013	0.021	0.019
	(2.390)	(2.309)	(1.954)	(2.650)	(−3.448)	(−1.182)	(−5.597)	(−3.657)	(4.100)	(2.490)	(5.054)	(4.367)
(-1,+30)	0.013	0.018	0.017	0.024	−0.036	0.018	−0.037	−0.009	0.050	0.000	0.054	0.033
	(2.188)	(3.098)	(3.146)	(4.388)	(−5.354)	(2.809)	(−6.329)	(−1.707)	(5.457)	(0.055)	(6.796)	(4.283)
(+2,+15)	0.003	0.000	0.016	0.017	−0.017	−0.002	−0.012	0.000	0.020	0.002	0.028	0.018
	(0.936)	(−0.001)	(3.509)	(3.823)	(−4.292)	(−0.529)	(−3.291)	(−0.116)	(3.819)	(0.389)	(4.802)	(3.028)
(+2,+30)	0.009	0.014	0.015	0.020	−0.029	0.020	−0.028	−0.003	0.038	−0.005	0.043	0.023
	(1.621)	(2.725)	(3.290)	(4.212)	(−4.652)	(3.365)	(−5.068)	(−0.633)	(4.603)	(−0.682)	(5.999)	(3.299)
N	700	700	563	563	961	961	1,343	1,343				

Table 6: Returns around announcement dates of buybacks or SEO pricings dates

This table reports returns around the announcements of buybacks and SEO pricings. The analysis is conducted for firms that announce buybacks or price an equity offering up to 2 days prior to the next earnings announcement. In addition to raw returns, four-factor (Fama-French Carhart) abnormal returns are reported. The t statistics are reported in parentheses below each estimate. N shows the number of observations. See Table 1 for the methodology used for calculating abnormal returns.

Day 0: Announcement date for buybacks or SEO pricings				
	Buybacks (Up to 2 days before earnings)		SEOs (Up to 2 days before earnings)	
Window	Four-factor	Raw	Four-factor	Raw
(-1,+1)	0.020	0.020	−0.020	−0.014
	(15.591)	(14.497)	(−21.154)	(−14.653)
(0,+1)	0.023	0.022	−0.013	−0.009
	(21.341)	(20.411)	(−15.079)	(−10.141)
(-1,+3)	0.025	0.025	−0.019	−0.010
	(17.476)	(16.337)	(−17.575)	(−9.355)
(-1,+5)	0.026	0.028	−0.020	−0.008
	(16.888)	(17.202)	(−17.265)	(−6.825)
(-1,+30)	0.042	0.065	−0.045	0.011
	(13.098)	(19.856)	(−21.051)	(5.487)
(+2,+15)	0.011	0.022	−0.008	0.015
	(6.736)	(12.628)	(−6.542)	(12.459)
(+2,+30)	0.022	0.046	−0.025	0.025
	(7.389)	(14.939)	(−13.733)	(14.689)
N	4,737	4,737	8,846	8,846

Table 7: Descriptive statistics for control variables

Summary statistics of the main variables used in constructing the regression models are reported below. Panel A presents summary statistics of various firm characteristics for repurchasing sample. BidAskSpread is the average bid-ask spread from 5 to 30 days before earnings announcement dates. Bid-ask spreads are estimated from daily high and low prices using the methodology in Corwin and Schultz (2012). ForecastError is the absolute difference between the actual quarterly earnings per share and the average of all the latest outstanding forecasts scaled by the share price at the beginning of the quarter. StdForecast is the standard deviation of all the latest outstanding forecasts scaled by the share price at the beginning of the quarter. Size is the market capitalization (in millions). FirmRet is the firm's return over a 1-month period before the buyback announcement while MktRet is the market return over the same period. M/B is the ratio of the market value to the book value of equity and FrequentDum is a dummy variable which takes a value of 1 for moderately or frequently repurchasing (issuing) firms. Panel B presents summary statistics of several control variables for SEOs sample where variables have the same definitions as above. To mitigate the effect of outliers, variables ForecastError, StdForecast, and M/B are winsorized at the 1% and 99% of their empirical distribution. The sample includes firms that announce buybacks or equity pricings up to 2 days before their earnings release date. For each variable, the number of non missing observations (N), the first quartile (Q1), mean, median, and the third quartile (Q3) are reported.

Panel A: Buybacks					
Variable	N	Q1	Mean	Median	Q3
BidAskSpread×100	4,569	0.582	1.340	0.972	1.636
ForecastError×100	3,444	0.025	0.573	0.095	0.291
StdForecast×100	2,911	0.026	0.212	0.063	0.150
Size ($Millions)	4,672	88.921	6844.239	350.985	2025.452
FirmRet (%)	4,734	−10.386	−4.240	−2.065	3.448
MktRet (%)	4,737	−3.238	−0.524	0.522	2.953
M/B	4,015	1.044	2.251	1.555	2.471
FreqentDum	4,737	0.000	0.323	0.000	1.000
Panel B: SEOs					
BidAskSpread×100	7,354	0.560	1.323	0.941	1.626
ForecastError×100	5,102	0.043	5.489	0.162	0.637
StdForecast×100	4,148	0.037	1.604	0.105	0.335
Size ($Millions)	8,835	157.151	2014.303	487.275	1511.176
FirmRet (%)	8,842	−5.934	1.578	0.385	7.091
MktRet (%)	8,844	−1.312	0.995	1.200	3.530
M/B	6,968	1.137	3.023	1.827	3.350
FreqentDum	8,844	0.000	0.420	0.000	1.000

Table 8: Determinants of returns: Buybacks sample

This table reports the estimates from regressing returns around earnings announcements on different sets of control variables. The dependent variable is the four-factor abnormal returns over the (+2,+15) and (+2,+30) windows. The analysis is done for the firms that announce a buyback program before their earnings release date. BidAskSpread is the average bid-ask spread from 5 to 30 days before earnings announcement dates. Bid-ask spreads are estimated from daily high and low prices using the methodology in Corwin and Schultz (2012). ForecastError is the absolute difference between the actual quarterly earnings per share and the average of all the latest outstanding forecasts scaled by the share price at the beginning of the quarter. StdForecast is the standard deviation of all the latest outstanding forecasts scaled by the share price at the beginning of the quarter. Size is the market capitalization (in millions). FirmRet is the firm's return over a 1-month period before the buyback announcement while MktRet is the market return over the same period. M/B is the ratio of the market value to the book value of equity and FrequentDum is a dummy variable which takes a value of 1 for moderately or frequently repurchasing firms. To mitigate the effect of outliers, ForecastError, StdForecast, and M/B are winsorized at the 1% and 99% of their empirical distribution. The analysis is conducted for firms that announce buybacks or price an equity offering up to 2 days prior to the next earnings announcement. Standard errors are reported below the coefficient estimates in parentheses. N shows the number of observations. Symbols ∗∗∗, ∗∗, and ∗ indicate significance at the 1%, 5%, and 10% levels, respectively.

	Four-factor: (+2,+15)				Four-factor: (+2,+30)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intercept	0.005	0.022***	0.026***	0.010	−0.011	0.027***	0.033***	0.004
	(0.007)	(0.006)	(0.008)	(0.010)	(0.010)	(0.010)	(0.012)	(0.015)
BidAskSpread	0.403**			0.681**	0.706***			1.325***
	(0.164)			(0.272)	(0.229)			(0.416)
Log(1+ForecastError)		0.140		0.570***		0.272*		0.544*
		(0.106)		(0.204)		(0.160)		(0.312)
Log(1+StdForecast)			1.067***	0.105			2.116***	1.296
			(0.391)	(0.556)			(0.598)	(0.852)
Log(Size)	−0.000	−0.002**	−0.003***	−0.001	0.001	−0.003**	−0.004***	−0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
FirmRet	−0.026**	−0.052***	−0.052***	−0.038**	−0.085***	−0.098***	−0.073***	−0.055**
	(0.012)	(0.013)	(0.015)	(0.015)	(0.017)	(0.020)	(0.022)	(0.023)
MktRet	0.007	0.026	0.018	0.011	0.008	−0.012	−0.039	−0.035
	(0.033)	(0.034)	(0.038)	(0.038)	(0.046)	(0.052)	(0.057)	(0.058)
M/B	0.001	0.001	0.001	0.001	0.003***	0.003***	0.003***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
FrequentDum	−0.005	−0.004	−0.002	−0.000	−0.002	−0.002	0.000	0.002
	(0.004)	(0.005)	(0.005)	(0.005)	(0.006)	(0.007)	(0.008)	(0.008)
Adjusted R^2	0.003	0.010	0.015	0.021	0.014	0.017	0.021	0.029
N	3,896	2,948	2,478	2,443	3,896	2,948	2,478	2,443

Table 9: Determinants of returns: SEOs sample

This table reports the estimates from regressing returns around earnings announcements on different sets of control variables. The dependent variable is the four-factor abnormal returns over the (+2,+15) and (+2,+30) windows. The analysis is done for the firms that price an equity offering before their earnings release date. BidAskSpread is the average bid-ask spread from 5 to 30 days before earnings announcement dates. Bid-ask spreads are estimated from daily high and low prices using the methodology in Corwin and Schultz (2012). ForecastError is the absolute difference between the actual quarterly earnings per share and the average of all the latest outstanding forecasts scaled by the share price at the beginning of the quarter. StdForecast is the standard deviation of all the latest outstanding forecasts scaled by the share price at the beginning of the quarter. Size is the market capitalization (in millions). FirmRet is the firm's return over a 1-month period before the equity offering while MktRet is the market return over the same period. M/B is the ratio of the market value to the book value of equity and FrequentDum is a dummy variable which takes a value of 1 for moderately or frequently issuing firms. To mitigate the effect of outliers, ForecastError, StdForecast, and M/B are winsorized at the 1% and 99% of their empirical distribution. The analysis is conducted for firms that announce buybacks or price an equity offering up to 2 days prior to the next earnings announcement. Standard errors are reported below the coefficient estimates in parentheses. N shows the number of observations. Symbols ∗∗∗, ∗∗, and ∗ indicate significance at the 1%, 5%, and 10% levels, respectively.

	Four-factor: (+2,+15)				Four-factor: (+2,+30)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intercept	−0.002	−0.014	−0.020**	0.008	−0.010	−0.031**	−0.047***	0.016
	(0.007)	(0.009)	(0.010)	(0.013)	(0.011)	(0.014)	(0.016)	(0.021)
BidAskSpread	−0.353***			−1.431***	−0.652***			−3.073***
	(0.104)			(0.327)	(0.160)			(0.531)
Log(1+ForecastError)		−0.020		0.006		−0.046*		−0.081*
		(0.015)		(0.030)		(0.024)		(0.049)
Log(1+StdForecast)			−0.015	−0.001			−0.012	0.159*
			(0.036)	(0.058)			(0.059)	(0.094)
Log(Size)	−0.003**	−0.002	−0.000	−0.003*	−0.004**	−0.003	0.000	−0.005*
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
FirmRet	−0.034***	−0.046***	−0.024*	−0.026**	−0.063***	−0.070***	−0.048**	−0.056***
	(0.010)	(0.011)	(0.012)	(0.013)	(0.015)	(0.018)	(0.020)	(0.021)
MktRet	0.005	0.064	0.042	0.031	0.017	0.076	0.041	−0.012
	(0.040)	(0.047)	(0.049)	(0.059)	(0.062)	(0.076)	(0.081)	(0.095)
M/B	0.000	0.000	−0.000	0.000	−0.000	0.000	−0.001	0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
FrequentDum	0.018***	0.014***	0.011**	0.013*	0.027***	0.022***	0.019**	0.019
	(0.004)	(0.005)	(0.005)	(0.007)	(0.006)	(0.008)	(0.009)	(0.011)
Adjusted R^2	0.007	0.005	0.001	0.007	0.009	0.005	0.002	0.014
N	5,772	4,029	3,306	2,777	5,772	4,029	3,306	2,777

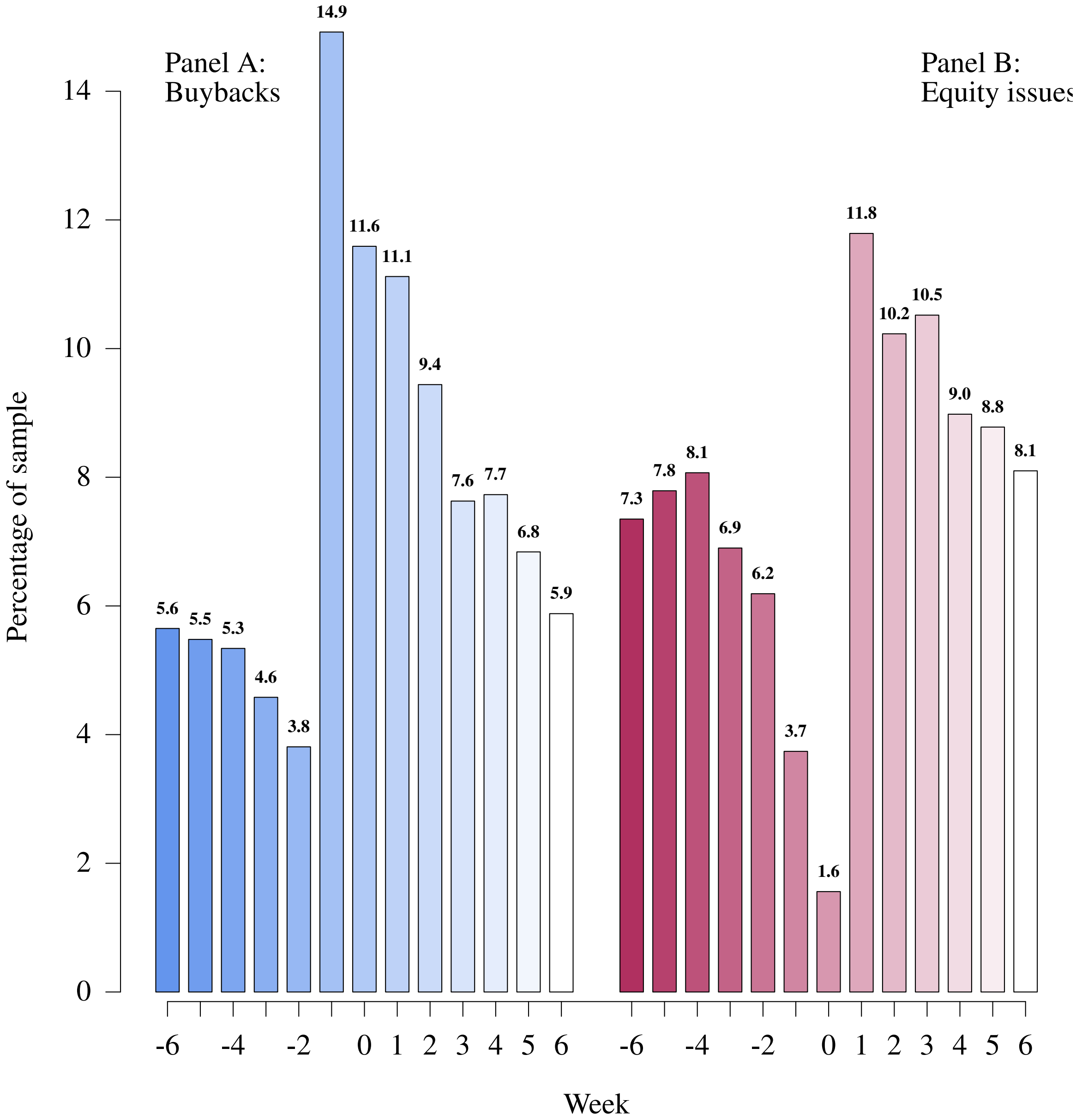

Figure 1: Distribution of buybacks and SEOs around earnings announcements

Panel A shows the histogram of repurchasing firms and Panel B depicts the histogram of issuing firms. Each bin in this graph represents 5 trading days. Positive sign indicates "after" and negative sign indicates "before" earnings announcement date. The bin labeled 0 indicates the event day which is the earnings announcement date.

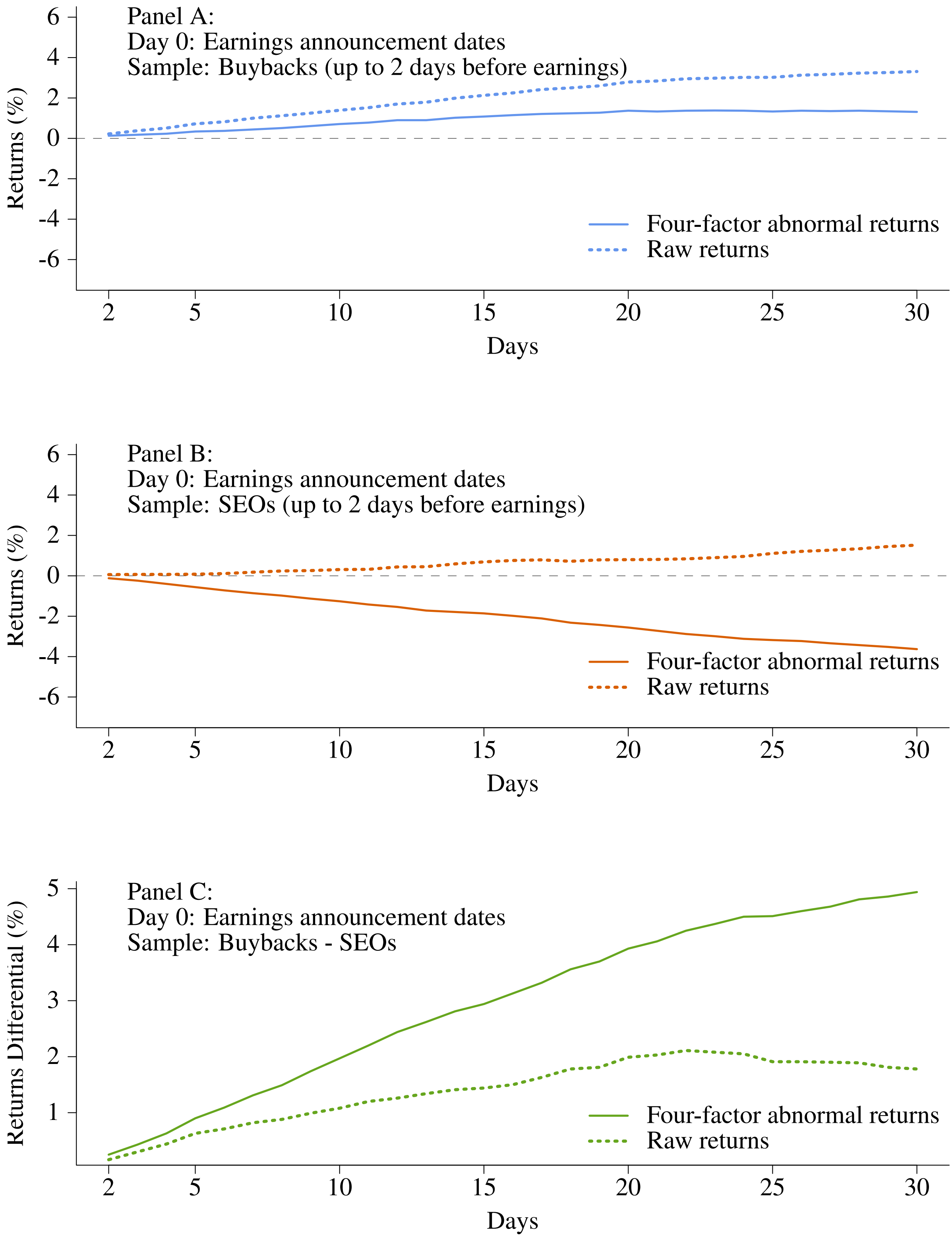

Figure 2: Returns around earnings for repurchasing and issuing firms

This figure shows cumulative average abnormal returns (CARs) and raw returns around earnings announcement. Panel A plots the returns around the earnings announcement date for repurchasing firms that made their buyback announcement up to 2 days before earnings. Panel B plots the returns around earnings announcement date for issuing firms that priced an equity offering up to 2 days before earnings. Panel C plots the difference between returns around earnings announcement dates for repurchasing firms and issuing firms. In all graphs, the returns are plotted from 2 days after to 30 days after earnings announcement.

Table A1: Summary statistics for repurchasing and issuing firms

This table presents the time profile and summary statistics for the sample of repurchasing firms over the 1994 to 2015 period and the sample of issuing firms over the 1970 to 2015 period. The second column reports the number of buybacks in each year, the third column states the median percent of shares outstanding to be repurchased, and the fourth column shows the median market cap of repurchasing firms. Similar information for equity issues is presented in columns 5 to 7. Market cap values are in 2013 dollars using CPI-U inflation data.

	Buybacks			SEOs		
Year	N	% of shares	Market cap	N	% of shares	Market cap
1970				105	8.66	1195
1971				271	11.39	664
1972				239	11.15	854
1973				146	10.78	918
1974				93	10.42	748
1975				189	11.42	897
1976				207	11.81	620
1977				145	13.14	534
1978				197	12.48	454
1979				172	12.87	416
1980				354	14.55	322
1981				397	13.34	308
1982				427	8.49	868
1983				775	14.47	327
1984				238	13.02	268
1985				382	17.22	250
1986				493	16.86	261
1987				298	17.06	283
1988				131	19.73	200
1989				223	19.34	203
1990				193	17.98	305
1991				477	17.54	345
1992				518	17.74	366
1993				659	20.57	324
1994	679	5.07	330	394	19.62	363
1995	693	5.07	229	572	20.02	400
1996	929	5.19	223	703	21.20	401
1997	815	5.34	291	681	18.71	517
1998	1377	5.44	232	553	13.77	883
1999	1093	5.45	239	442	15.95	981
2000	945	5.77	213	409	11.30	1692
2001	830	5.10	179	397	14.04	962
2002	656	5.12	269	377	13.34	754
2003	494	5.10	531	450	13.20	806
2004	570	5.04	1182	513	13.93	835
2005	660	5.20	1346	403	15.65	804
2006	647	5.31	1228	450	14.50	906
2007	813	6.00	877	389	14.87	867
2008	749	7.16	354	266	11.14	1337
2009	269	6.00	463	801	12.61	433
2010	423	6.14	1351	754	12.54	415
2011	542	6.15	1407	624	12.86	468
2012	430	5.95	1186	721	11.97	565
2013	426	5.30	1478	842	12.29	646
2014	517	5.26	1711	726	11.92	705
2015	549	6.06	1293	670	12.40	937
Total	15,106	5.38	481	19,466	14.22	551

Table A2: Distribution of buybacks and SEOs

This table reports the size of samples used for different analyses. Panel A shows the total sample size. Panel B shows the number of buybacks or SEO pricings which are closest to but prior to the earnings announcements. Panel C shows the number of buybacks or SEO pricings which are closest to but 2 or more days prior to the earnings announcements. Panel D shows the number of buybacks or SEO pricings which are closest to but 2 or more days prior to the earnings announcements and have a market cap greater than \$100 million. Panel E shows the number of buybacks or SEO pricings which are closest to but 2 or more days prior to the earnings announcements and are not financial or utility firms. Panel F shows the number of buybacks or SEO pricings which are closest to but 2 or more days prior to the earnings announcements and are greater than 5% of shares outstanding for repurchasing firms and greater than 10% of shares outstanding for issuing firms. Panels G, H, I, and J show the number of buybacks or SEO pricings which are closest to but 2 or more days prior to the earnings announcements and are announced between 1994 and 2000, 2001 and 2005, 2006 and 2010, and 2011 and 2015, respectively.

	Buybacks	SEOs
Panel A: Total number of observations	15,106	19,466
Panel B: Number of observations closest to but prior to the earnings announcemenets	6,026	9,148
Panel C: Number of observations closest to but 2 or more days prior to the earnings announcemenets	4,737	8,846
Panel D: Number of observations closest to but 2 or more days prior to the earnings announcemenets with market cap>\$100 million	3,407	7,299
Panel E: Number of observations closest to but 2 or more days prior to the earnings announcemenets excluding financial and utility firms	2,760	5,793
Panel F: Number of observations closest to but 2 or more days prior to the earnings announcemenets with buybacks>5% and SEOs>10%	2,741	5,976
Panel G: Number of observations closest to but 2 or more days prior to the earnings announcemenets between 1994 and 2000	2,434	1,679
Panel H: Number of observations closest to but 2 or more days prior to the earnings announcemenets between 2001 and 2005	1,040	815
Panel I: Number of observations closest to but 2 or more days prior to the earnings announcemenets between 2006 and 2010	700	961
Panel J: Number of observations closest to but 2 or more days prior to the earnings announcemenets between 2011 and 2015	563	1,343